

Lecture 3
Under-determination
Flipped classroom:
online lecture preceding
Tuesday, 15 September 2026

PHILOSOPHY OF SCIENCE





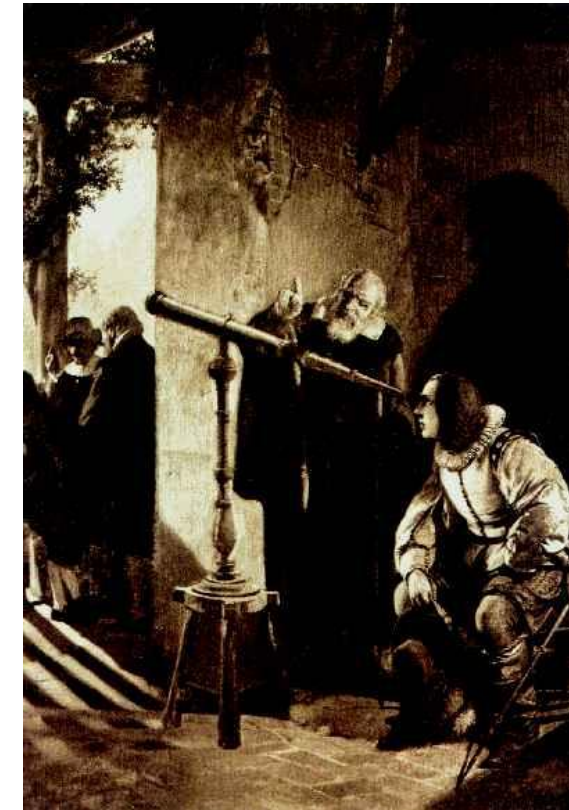
OUTLINE

- (A) Duhem's under-determination
- (B) Quine's Two Dogmas
- (C) Critiques of Quine
- (D) Empirical Under-determination

(A) DUHEM'S
UNDER-
DETERMINATION

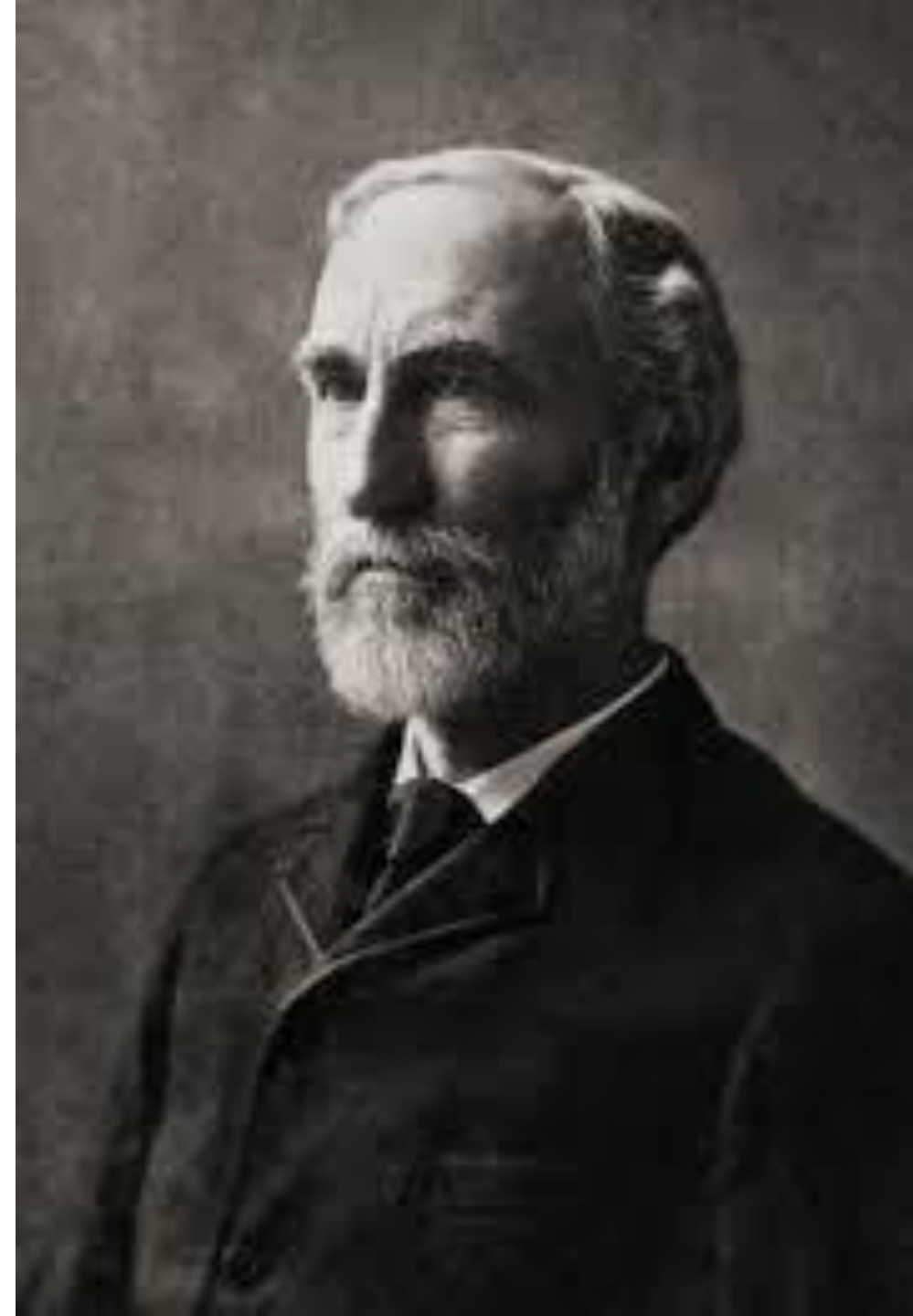
THEORIES REFUTED BY OBSERVATIONS? THE EMPIRICAL BASIS

- Problem: Suppose an astronomical theory conflicts with observations made with telescopes
 - Which theory is refuted: astronomical theory or theory of optics?
- Popper's solution: There are no objective facts that determine our choice. We have to accept an empirical basis by **intersubjective agreement** (consensus)
 - Acceptance of observation report that falsifies theory is a *human decision*



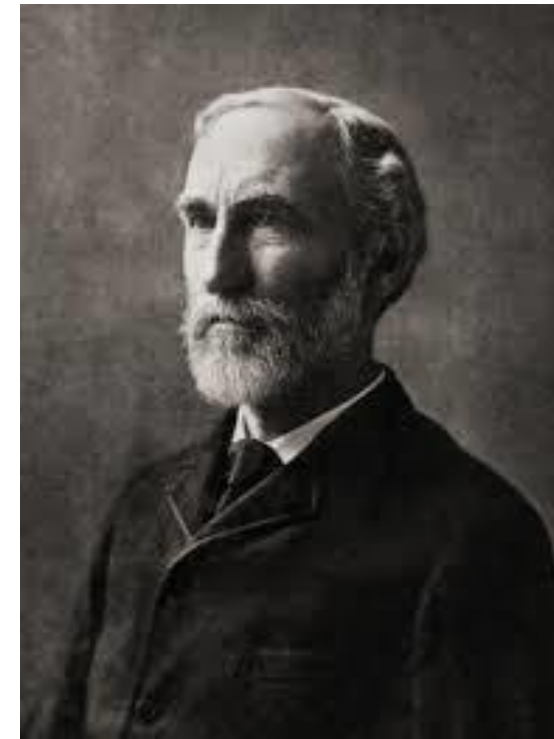
DUHEM'S UNDER-DETERMINATION

- Given that a set of premises $\{P_1, P_2, \dots, P_n\}$ deductively entails a statement O describing a possible observation, and that from experimental observation one concludes that the statement O' is true, where O' entails that O is false, it follows that the conjunction of the premises is false. It does not follow that any particular premise is false.
 - Falsification is ambiguous



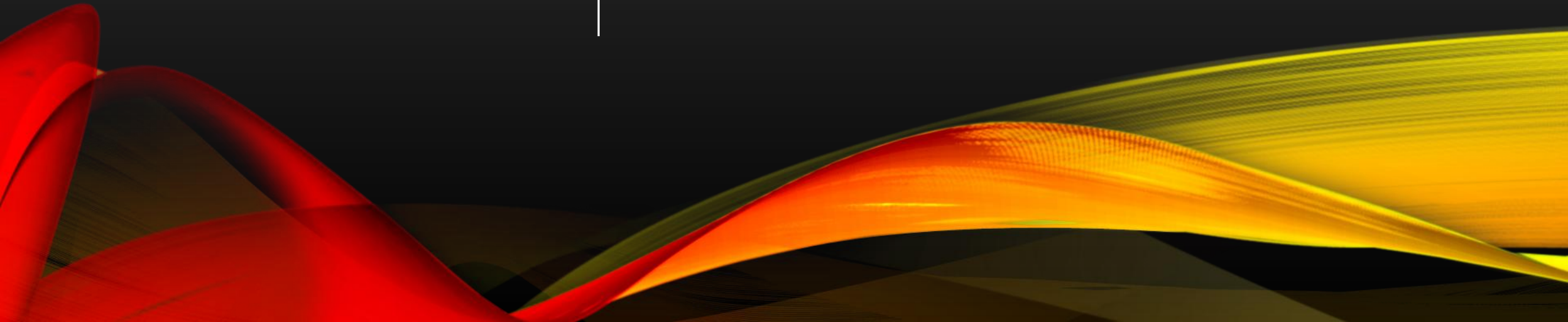
PIERRE DUHEM, UNDER-DETERMINATION AND GOOD SENSE

- **Crucial experiment:** ‘an irrefutable procedure for transforming one of the two hypotheses before us into a demonstrated truth’.
 - (Notion of ‘experimentum crucis’ goes back to Francis Bacon, Robert Hooke, Newton)
- Impossible, because hypotheses only tested in combination with auxiliary assumptions (e.g. experimental apparatus)
 - There may be **unconceived alternatives** (cf. Stanford)
- Scientists must use their ‘**good sense**’: ‘certain opinions which do not fall under the hammer of the principle of contradiction are in any case perfectly unreasonable’.
 - What might good sense be?
 - How can a scientist be ‘an impartial and faithful judge?’



The analytic/synthetic
distinction
Quine's "two dogmas"
The "web of belief"

(B) QUINE'S "TWO DOGMAS"



W.V.O. QUINE: TWO DOGMAS OF EMPIRICISM

1951 paper attacks two dogmas of the dominant logical-positivist view of science:

1. **Reductionism:** Scientific statements are logically equivalent to statements in special empiricist language.
 - Subjective language of immediate experience or objective language of physical things.
 - Verification theory of meaning.
 2. **Analytic-synthetic distinction:** 'difference between conceptual schemes and facts is of degree only'
- Quine: at bottom, the two dogmas are identical
 - Analytic statements are limit of true statements:
 - No empirical content; confirmed no matter how the world is



W.V.O. QUINE: TWO DOGMAS OF EMPIRICISM

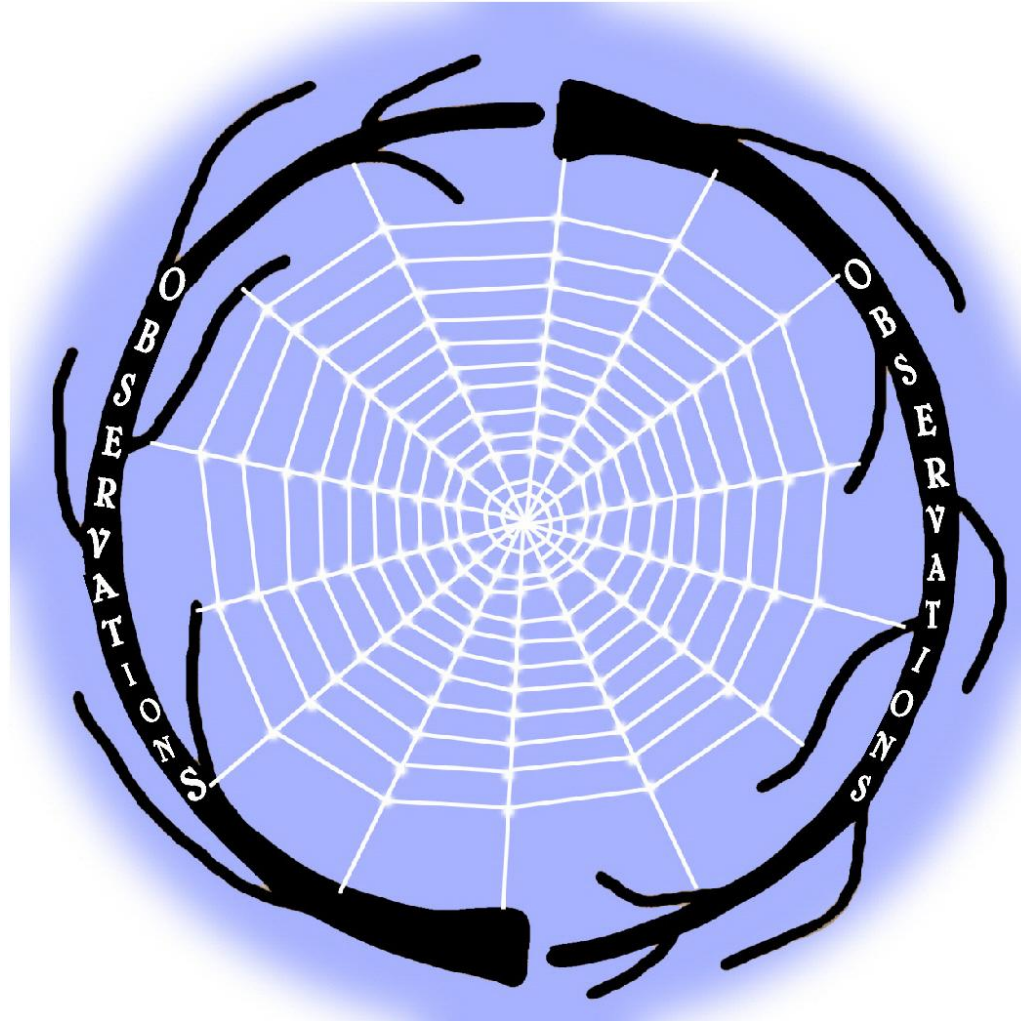
1951 paper attacks two dogmas of the dominant logical-positivist view of science:

1. **Reductionism:** verification theory of meaning
Cannot test isolated statements
2. **Analytic-synthetic distinction:** 'difference between conceptual schemes and facts is of degree only'

➔ The whole of our knowledge is a 'web of belief' that is under-determined by experience



QUINE'S WEB OF BELIEF





THE WEB OF BELIEF AND UNDER-DETERMINATION

- *The totality of our so-called knowledge or beliefs, from the most casual matters of geography and history to the profoundest laws of atomic physics or even of pure mathematics and logic, is a man-made fabric which **impinges on experience only along the edges**. Or, to change the figure, total science is like a field of force whose **boundary conditions are experience**. A **conflict with experience at the periphery occasions readjustments in the interior** of the field.*
- But the total field is so **underdetermined by its boundary conditions**, experience, that there is much latitude of choice as to what statements to reevaluate in the light of any single contrary experience.



THE DUHEM-QUINE THESIS

- One way to interpret the web is in terms of alternative **hypotheses/theories**
- Disconfirming an observation only affects the web “at its edges”
- Auxiliary theories/hypotheses
 - See Laudan’s discussion of Quine’s “four stratagems”

(C) CRITIQUES OF QUINE

Howard Stein and Larry Laudan

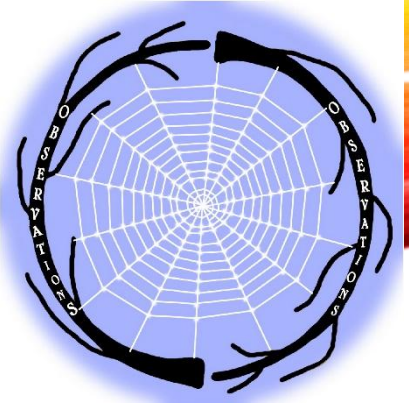


CRITIQUE OF QUINE

- At heart of problem: **Quine's scepticism about meanings.**
- **Rejects outright various theories of meaning.** Frege and Carnap distinguish two types of meanings:
 - **Sense (intension):** linguistic meaning of the words. *Morning/evening star.*
 - **Reference (extension):** actual object/entity the words refers to. *Venus.*
- **Standard account of intension** of sentence S in possible world semantics:
 - Maximal set of set of possible worlds where S is true.
 - Alternatively: map from possible worlds to truth values where S true/false.
- Attributes to Leibniz for analyticity, writes off as '*picturesque*'.
 - Long discussion intensions as remnants of Aristotelian essences.
- Assumption: empiricist is sceptical about meanings.
 - Intensions would have merited more detailed engagement.
 - Long discussion engages little with meanings in this sense.

LAUDAN ON UNDER-DETERMINATION

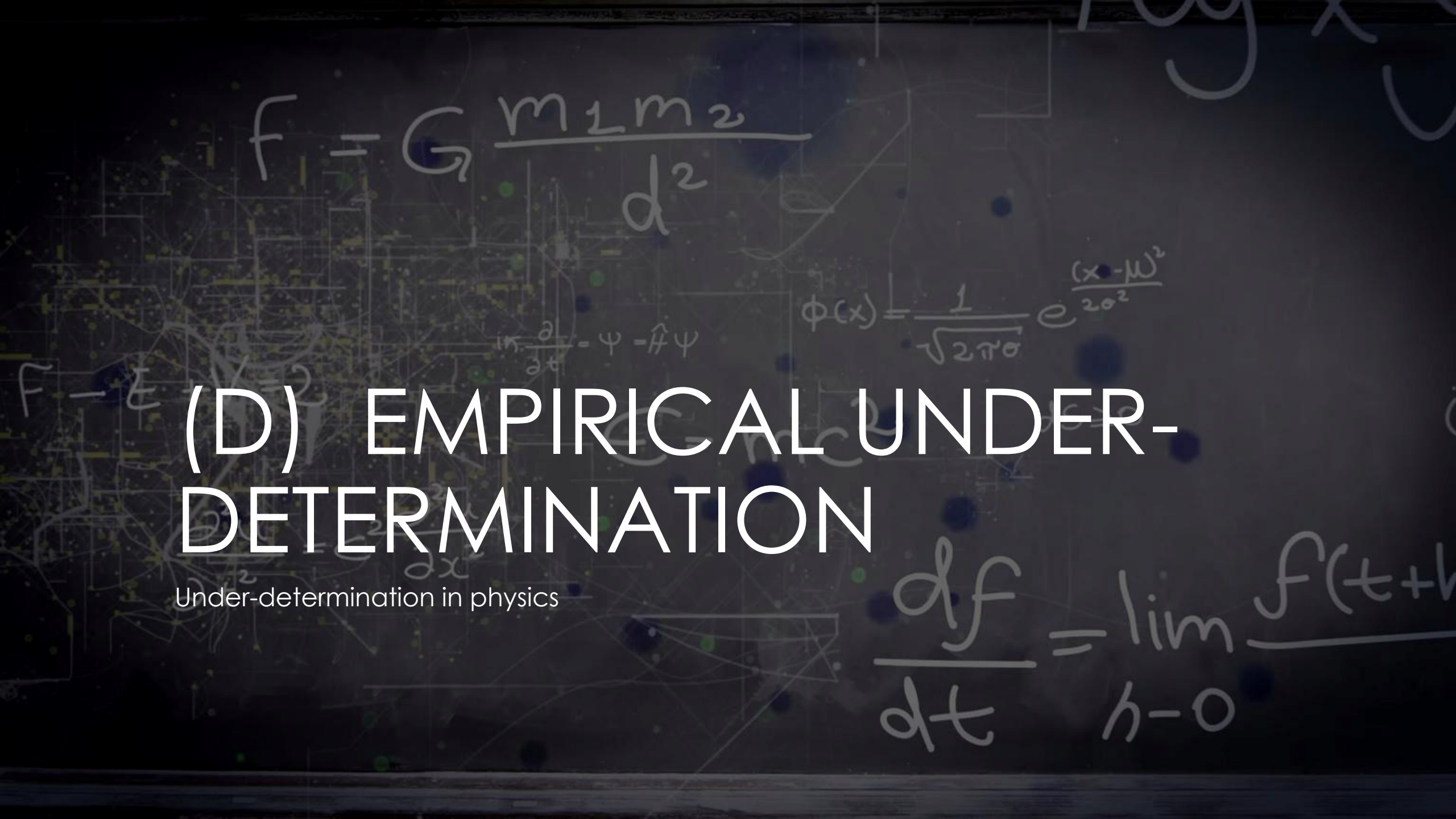
- 'Sloppy formulations of the thesis of UD have encouraged authors to use it ... to support whatever relativist conclusions they fancy' (p. 289).
- **Logic and rationality:** 'Too many of the discussions of UD in the last quarter century have proceeded in an evaluative vacuum. They imagine that if a course of action is **logically possible**, then one need not attend to the question of its **rationality**' (p. 298).
 - Rationality includes epistemic warrant/justification: especially the empirical evidence.



QUINE'S WEB OF BELIEF

Quine: Any statement can be held true come what may, if we make drastic enough adjustments elsewhere in the system. Even a statement very close to the periphery can be held true in the face of recalcitrant experience by pleading hallucination or by amending certain statements of the kind called logical laws (p. 297).

Laudan: What grounds does Quine have for asserting [this]? One might expect that he could establish [its] plausibility ... in virtue of examining the relevant rules of rational theory choice and showing, if it could be shown, that those rules were always so ambiguous that, confronted with any pair of theories and any body of evidence, they could never yield a decision ... But Quine *nowhere* ... engages in a general examination of ampliative rules of theory choice (p. 293)

The background is a dark, textured surface resembling a chalkboard, filled with faint, handwritten mathematical equations in white and yellow. Visible equations include Newton's law of gravitation $F = G \frac{m_1 m_2}{d^2}$, the Schrödinger equation $i\hbar \frac{\partial}{\partial t} \psi = \hat{H} \psi$, a Gaussian function $\phi(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$, and a derivative $\frac{df}{dt} = \lim_{h \rightarrow 0} \frac{f(t+h) - f(t)}{h}$.

(D) EMPIRICAL UNDER-DETERMINATION

Under-determination in physics



EMPIRICAL UNDER-DETERMINATION

- Quine (1975 *On empirically equivalent systems of the world*): imagine two theory formulations that are:
 - (1) **Empirically equivalent**: same observation sentences
 - (2) **Theoretically inequivalent**: different theoretical sentences
- **Theory under-determined by the empirical data**: empirical data do not determine a choice.
- Seen as a problem for **scientific realism**: cannot decide which theory to be realists about.
- Two cases:
 - **Semantic**: regards meaning/interpretation (regardless of evidence)
 - **Epistemic**: regards the empirical evidence

EXAMPLES OF EMPIRICAL UNDER-DETERMINATION?

- Laudan and Leplin: It is noteworthy that **contrived examples alleging empirical equivalence always involve the relativity of motion**; it is the impossibility of distinguishing apparent from absolute motion to which they owe their plausibility. This is also the problem in the pre-eminent historical examples, the competition between Ptolemy and Copernicus, which created the idea of empirical equivalence in the first place, and that between Einstein and Lorentz.
- Quine (1975): whether any cases exist is an 'open question'
- Stanford (2006): The critics of underdetermination have been well within their rights to **demand that serious, nonsceptical, and genuinely distinct empirical equivalents** to a theory actually be produced before they withhold belief in it and refusing to presume that such equivalents exist when none can be identified.



EARLY EXAMPLE: POINCARÉ'S CONVENTIONALISM

- **Non-Euclidean (curved) geometries** mathematically possible/consistent: Gauss, Riemann, 19th century.
- Kant: geometry and Newton's three laws are synthetic *a priori*, i.e. **independent of experience**.
 - A priori, but they add something to experience: not mere logical truths.
 - Specific forces e.g. inverse-square law and their parameters are empirical, i.e. synthetic a posteriori.
- Poincaré (*Science and Hypothesis* 1902): geometry **is neither purely a priori nor a posteriori**: it is **conventional**.
 - We choose the geometric viewpoint that explains experience in the simplest way, but could take alternative viewpoints.

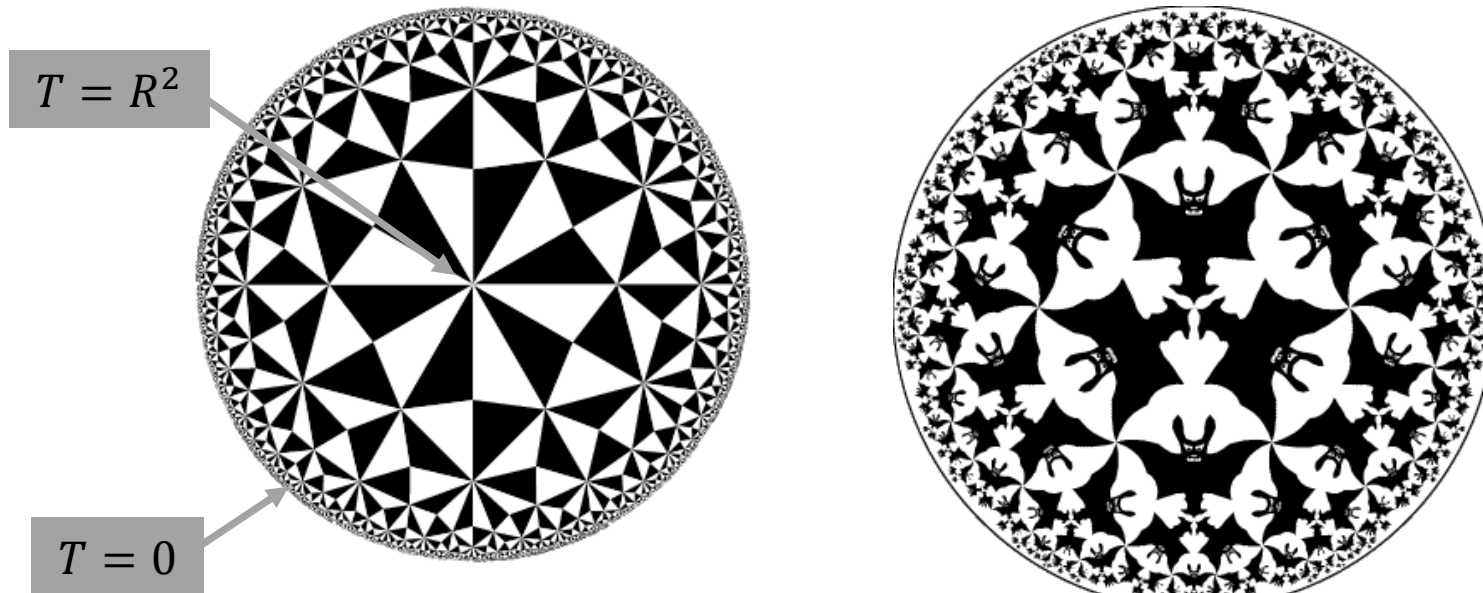


POINCARÉ'S CONVENTIONALISM

- First argument: analysis of **perception**.
- We **don't perceive space directly as three-dimensional**:
 - Our field of vision is prima facie two-dimensional, derivatively 3D
 - Our "tactile space" is (could be) multi-dimensional
- By experimentation we "discover" the structure of space, but this structure is **not unique**.
- Second argument: a thought experiment:

POINCARÉ'S THOUGHT EXPERIMENT

- Two versions of a world inside a sphere/disk of radius R :
 1. A world with a **temperature**, $T = R^2 - r^2$, highest at the centre of the sphere, radially decreasing to zero. If all bodies have the same coefficient of expansion proportional to T , then the size of any body is highest at the centre, and decreases to zero at the boundary.
 2. Beings whose bodies have the same **coefficient of expansion**, will experience the space as infinite, and curved: if they develop a geometry describing their world, it will be non-Euclidean.



POINCARÉ'S CONVENTIONALISM

- For us, beings whose sensory organs are accustomed to experiencing Euclidean space, it appears easier to describe this world as an **Euclidean sphere/disk**; a law of physics dictates that objects expand with temperature.
- No fact of the matter which of the two descriptions is right: our mental habits favour Euclidean picture, but beings just like us that happened to live inside Poincaré's sphere would describe it in terms of **non-Euclidean geometry**.
- No experiment can decide for one or the other, since every experimental set-up can itself be interpreted in either mental picture.
- The choice is made by **convention**, similarly to how we can measure distances in metres or in yards.
- Duhem: laws cannot be tested directly, but can be tested *indirectly*



CONCLUSION

- Duhem: under-determination resolved by “good sense”
- Quine is sceptical of ‘meanings’: undercut of non-empirical knowledge
- **Web of belief** against both dogmas
 - Rejects verification principle
- **Empirical under-determination:** theory under-determined by empirical data
 - Very common in scientific practice (*transient* under-determination)
- **Dualities** potential cases of semantic-under-determination
 - Not a threat to cautious scientific realism

FURTHER READING

- Curd & Cover, Commentary to Chapter 3

Empirical under-determination

- Quine, W. V. O. (1975), 'On Empirically Equivalent Systems of the World'.

Under-determination and duality:

- De Haro, S. (2021). 'The Empirical Under-Determination Argument against Scientific Realism for Dual Theories'.
- Le Bihan, B. and Read, J. (2018). 'Duality and Ontology'.